

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Department of Aerospace Engineering

Research Internship Report

Vector Field UAV Guidance for Path Following and Obstacle Avoidance

Singularity-Free Vector-Field-Based Path Following with an Exploratory
Obstacle-Avoidance Extension

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Internship Period: June–July 2026

September 2026

Declaration

I declare that this report presents the work carried out during my research internship at the Unmanned Vehicle Systems Lab, Department of Aerospace Engineering, Indian Institute of Technology Kharagpur, under the supervision of Prof. Sikha Hota.

The central singularity-free vector-field construction is based on the published work of Yao et al. [1]. The implementation and engineering extensions described in this report were developed as part of the internship project. The distinction between material inherited from the reference work and project-specific implementation, numerical, guidance, and exploratory extensions is stated explicitly throughout the report.

Dheeraaj
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Certificate

This is to certify that **Dheeraj** (22AE30009), an undergraduate student of the Department of Aerospace Engineering, Indian Institute of Technology Kharagpur, has successfully completed a research internship at the **Unmanned Vehicle Systems Lab** under my supervision.

During the internship, he worked on the research project titled “**Vector Field UAV Guidance for Path Following and Obstacle Avoidance**”. The research work was carried out during the months of **June and July 2026** and involved the application of concepts and techniques related to UAV guidance, path following, and obstacle avoidance.

Date: September 15, 2026

Place: IIT Kharagpur

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Abstract

Vector-field-based path following is an attractive guidance strategy for autonomous aerial vehicles because it provides a continuous direction field over the workspace and separates the geometric path-following problem from the vehicle’s lower-level control dynamics. A fundamental difficulty, however, appears when the desired path is simple closed or self-intersecting. A time-invariant guiding vector field defined in the same dimensional space as the physical path cannot, in general, provide global convergence without singular points. At a singular point the vector field vanishes, and consequently its direction—and any normalized guidance law based on it—becomes undefined.

This internship investigated a higher-dimensional singularity-free vector-field formulation based primarily on the work of Yao, Garcia de Marina, Lin, and Cao [1]. For a parameterized planar path $x = f_1(w)$ and $y = f_2(w)$, an additional virtual coordinate w is introduced so that the physical path is represented as the lifted curve $\gamma(w) = [f_1(w), f_2(w), w]^\top$ in $\mathbb{R}^3$. The extra dimension separates distinct visits to the same physical point and changes the topology of the lifted path from a potentially closed or self-intersecting curve to an unbounded, non-self-intersecting curve. A guiding vector field is then constructed from the gradients of two implicit path constraints.

The work moved beyond a purely theoretical construction and developed a MATLAB simulation framework for practical UAV guidance. The implementation includes systematic initialization of the virtual path parameter, an adaptive Chebyshev-based root solver, smooth nonlinear convergence terms, a course-distribution guidance layer, wind-aware fixed-airspeed motion, course-rate and curvature constraints, explicit handling of degenerate physical projections, and synchronized 2D/3D visualization and diagnostics. The main simulation uses a self-intersecting figure-eight path and a fixed-airspeed vehicle under a configurable wind field.

An exploratory obstacle-avoidance extension was also investigated using vector-field modulation around implicit obstacles. Although obstacle avoidance was an intended extension of the singularity-free path-following framework, the attempted approach did not result in a satisfactory integrated and validated path-following/obstacle-avoidance solution. It is therefore treated in this report as an experimental investigation rather than as a successful final contribution. This negative result is useful because it exposes the additional difficulty of preserving both the singularity-free path-following structure and reliable obstacle-induced flow behavior within one guidance field.

The principal outcome of the internship is therefore a research-oriented implementation and analysis framework for singularity-free vector-field path following, together with a clear identification of the remaining problems required for a robust integrated obstacle-avoidance architecture.

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Nomenclature

Symbol	Meaning
ξ	Augmented state vector
x, y	Physical planar coordinates
w	Virtual/path-parameter coordinate
$\gamma(w)$	Higher-dimensional parameterized path
ϕ_i	Implicit constraint functions defining the lifted path
e	Vector of path-following errors
N	Matrix of constraint gradients
K	Positive diagonal convergence-gain matrix
$\mathbf{V}$	Augmented guiding vector field
$\mathbf{V}_{\text{phy}}$	Physical (x, y) projection of the guiding field
χ_d	Desired course angle
χ	Current course angle
e_χ	Wrapped course-angle error
$\dot{\chi}_{cmd}$	Commanded course rate
V_a	Airspeed magnitude
$\mathbf{W}$	Wind velocity
$\mathbf{V}_g$	Ground velocity
κ_{xy}	Horizontal-plane curvature
$\phi_{\max}$	Maximum allowable bank angle
g	Gravitational acceleration
$\mathcal{D}$	Degenerate physical-projection set

Chapter 1

Introduction

1.1 Motivation

Unmanned Aerial Vehicles (UAVs) have emerged as a critical technology for a wide range of applications, including surveillance, environmental monitoring, infrastructure inspection, and target tracking [2, 3, 4, 5]. Many of these applications require a UAV to autonomously follow a desired path while maintaining reliable and safe operation in environments that may contain disturbances, obstacles, and restricted regions. Consequently, the development of reliable guidance and navigation algorithms capable of generating continuous and dynamically feasible motion commands remains an important problem in autonomous aerial systems.

Traditionally, UAV guidance systems have relied heavily on waypoint-based navigation and trajectory-tracking approaches, in which a desired motion is represented by a sequence of spatial points or a time-parameterized trajectory generated offline. The vehicle is then commanded to track these waypoints or trajectory states using a control law designed for the specific vehicle platform [6, 7, 8]. Such approaches are widely used in practical autopilot architectures, including systems based on platforms such as Pixhawk and Piccolo. However, waypoint-based navigation may introduce several practical limitations. Discretization of a continuous path into a finite set of waypoints can result in increased cross-track error, particularly in the presence of disturbances such as wind, while aggressive waypoint transitions may impose demands that are inconsistent with the vehicle's kinematic and dynamic constraints.

For applications in which the desired motion is fundamentally geometric, a *path-following* formulation provides an attractive alternative to direct trajectory tracking. In path following, the objective is not to reproduce a prescribed position as a function of time, but rather to converge to and subsequently move along a desired geometric path. This distinction is particularly relevant for autonomous aircraft, since many practical missions require a vehicle to follow a spatial curve while allowing the along-path speed to vary according to vehicle dynamics, environmental conditions, and guidance requirements.

Among the various path-following techniques, Vector-Field-Guided Path Following (VF-PF) is particularly attractive because it provides a continuous desired direction at every point in the operating domain. A guiding vector field can be designed to simultaneously provide a component that drives the vehicle toward the desired path and a tangential component that propagates the vehicle along the path. Thus, unlike waypoint-based methods, the guidance command is generated continuously from the geometry of the desired path rather than only at a discrete set of reference points.

Despite these advantages, conventional vector-field-based path-following methods encounter

a fundamental limitation when the desired path possesses certain topologies. In particular, simple closed paths and self-intersecting paths cannot, in general, admit a globally convergent, time-invariant guiding vector field in the same-dimensional physical space without singular points. A singular point is a point at which the guiding vector field becomes zero, resulting in the loss of a unique guidance direction and rendering normalization of the vector field undefined. For self-intersecting paths, this difficulty is especially evident because multiple branches of the path may pass through the same physical point while possessing different tangent directions. The rigorous topological analysis presented by Yao *et al.* demonstrates that global convergence to simple closed or self-intersecting desired paths cannot be guaranteed using conventional same-dimensional time-invariant vector-field path-following formulations [1].

To overcome this limitation, Yao *et al.* proposed a higher-dimensional formulation in which the physical desired path is transformed into an unbounded and non-self-intersecting path by introducing an additional virtual coordinate [1]. For a planar parameterized path

$$x = f_1(w), \quad y = f_2(w), \quad (1.1)$$

the corresponding augmented path can be represented as

$$\gamma(w) = \begin{bmatrix} f_1(w) \\ f_2(w) \\ w \end{bmatrix}. \quad (1.2)$$

The additional coordinate distinguishes different occurrences of the same physical position and thereby removes the ambiguity associated with self-intersections. A guiding vector field can then be constructed in the higher-dimensional space without the singularities associated with the original physical representation. The reference work further establishes theoretical convergence properties of the resulting system and demonstrates its applicability to complex robotic and fixed-wing navigation problems [1].

The present research internship builds upon this higher-dimensional singularity-free vector-field formulation with the objective of investigating its implementation for UAV-oriented path following. The primary focus of the work was the development and simulation of singularity-free vector-field-based path following for parameterized planar curves, together with the numerical and guidance mechanisms required to connect the theoretical vector field to a practical vehicle model. In addition to the core vector-field construction, the implementation considers virtual-parameter initialization, numerical root finding, wind-aware motion, finite turning capability, curvature constraints, physical projection of the augmented vector field, and visualization of the resulting trajectories in both physical and augmented spaces.

A secondary objective of the internship was to investigate the incorporation of obstacle avoidance into the singularity-free path-following framework. The intention was to modify the guidance field near obstacles while retaining convergence toward the desired path away from them. An exploratory vector-field-modulation approach was therefore implemented and investigated. However, the obstacle-avoidance extension did not result in a satisfactory integrated solution with corresponding convergence and safety guarantees. Accordingly, the obstacle-avoidance component is presented in this report as an exploratory investigation and as a basis for future research, while the primary validated focus of the internship remains the singularity-free vector-field-based path-following framework.

1.2 Objectives

The internship objectives were:

1. Study the singularity problem in conventional same-dimensional vector-field path following.
2. Implement the higher-dimensional singularity-free formulation of Yao et al. [1].
3. Adapt the formulation to parameterized planar paths using an explicit virtual coordinate.
4. Develop robust numerical initialization of the virtual path parameter.
5. Build a practical guidance layer capable of producing course commands under wind and turning constraints.
6. Analyze the distinction between a singularity-free augmented field and a degenerate physical projection.
7. Develop 2D/3D visualization and quantitative diagnostics.
8. Investigate, but not overstate, an obstacle-avoidance extension based on vector-field modulation.

1.3 Scope of the Work

The final implementation is primarily a MATLAB simulation and guidance study. It is not presented as a flight-tested six-degree-of-freedom aircraft controller. The repository documentation explicitly characterizes the implementation as a research/simulation framework with a guidance/kinematic vehicle model rather than a production flight-control system.

Chapter 2

Background and Literature Foundation

2.1 Vector-Field Guided Path Following

Let a desired path $\mathcal{P} \subset \mathbb{R}^n$ be represented geometrically, and consider the autonomous system

$$\dot{\boldsymbol{\xi}} = \chi(\boldsymbol{\xi}), \quad (2.1)$$

where $\chi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a continuously differentiable guiding vector field.

The VF-PF problem can be stated informally as finding χ such that trajectories starting away from the path converge to it, trajectories initialized on the path remain on it, and the vector field does not vanish on the desired path. These requirements are important because the zero vector has no unique direction and therefore cannot provide a meaningful propulsion or course command.

For an implicit two-dimensional path

$$\mathcal{P} = \{(x, y) : \phi(x, y) = 0\}, \quad (2.2)$$

a classical construction uses a tangential component obtained by rotating the gradient and a normal component that reduces the path error. In matrix form, the 90° rotation is

$$E = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \quad (2.3)$$

A representative field is

$$\chi(x, y) = E\nabla\phi(x, y) - k\phi(x, y)\nabla\phi(x, y), \quad (2.4)$$

with $k > 0$. The first term is tangent to the level set $\phi = 0$, whereas the second term acts to reduce the constraint error.

The reference paper generalizes this construction to arbitrary dimensions by representing a one-dimensional desired path as the intersection of $n - 1$ hypersurfaces [1].

2.2 General Guiding Vector Field

Suppose

$$\mathcal{P} = \{\boldsymbol{\xi} \in \mathbb{R}^n : \phi_i(\boldsymbol{\xi}) = 0, i = 1, \dots, n - 1\}. \quad (2.5)$$

Define

$$N(\boldsymbol{\xi}) = \begin{bmatrix} \nabla\phi_1(\boldsymbol{\xi}) & \cdots & \nabla\phi_{n-1}(\boldsymbol{\xi}) \end{bmatrix}, \quad (2.6)$$

and

$$e(\boldsymbol{\xi}) = \begin{bmatrix} \phi_1(\boldsymbol{\xi}) & \cdots & \phi_{n-1}(\boldsymbol{\xi}) \end{bmatrix}^T. \quad (2.7)$$

Let

$$K = \text{diag}(k_1, \dots, k_{n-1}), \quad k_i > 0. \quad (2.8)$$

The reference field can be written compactly as

$$\boxed{\chi(\boldsymbol{\xi}) = \nabla \times \phi(\boldsymbol{\xi}) - N(\boldsymbol{\xi})Ke(\boldsymbol{\xi})} \quad (2.9)$$

where $\nabla \times \phi$ denotes the generalized cross product of the $n - 1$ constraint gradients.

The generalized cross product is orthogonal to every gradient:

$$\nabla \times \phi \perp \nabla\phi_i, \quad i = 1, \dots, n - 1. \quad (2.10)$$

Consequently, the first term is tangent to every constraint surface and therefore provides the propagation direction, while the second term pushes the state towards the intersection of the constraint surfaces.

2.3 Singular Points

A singular point of the vector field satisfies

$$\chi(\boldsymbol{\xi}_s) = 0. \quad (2.11)$$

At such a point the normalized field

$$\hat{\chi} = \frac{\chi}{\|\chi\|} \quad (2.12)$$

is not defined. This matters directly for guidance systems that extract heading/course from a normalized vector.

The crucial result of Yao et al. is that, for simple closed or self-intersecting desired paths, this difficulty is not removable merely by changing controller gains. It follows from a topological obstruction associated with a time-invariant vector field in the same dimensional space as the path [1].

Chapter 3

The Topological Singularity Problem

3.1 Self-Intersecting Paths

Consider a self-intersecting planar curve represented implicitly by

$$\mathcal{P} = \{(x, y) : \phi(x, y) = 0\}. \quad (3.1)$$

At a crossing point, two different local branches occupy the same physical point but have different tangent directions.

For the field in (2.4), the propagation term is proportional to a tangent vector obtained from $\nabla\phi$. At a transverse crossing, the implicit representation cannot provide a single regular one-dimensional manifold structure at the intersection. The paper proves that such crossing points must belong to the singular set for the conventional construction.

Proposition 3.1 (Singularity at a self-intersection). *For a self-intersecting desired path represented by the intersection of the required hypersurfaces, the conventional same-dimensional guiding vector field necessarily has at least one singular point on the path.*

Proof. Assume, for contradiction, that a self-intersection point ξ_s is regular for the implicit description, i.e. the constraint gradients are linearly independent there. By the implicit-function theorem, the zero set of the constraints must then be locally a unique one-dimensional embedded manifold near ξ_s . Such a regular manifold has a unique local tangent direction. A genuine self-intersection, however, contains at least two distinct local branches and therefore two distinct tangent directions through the same point. This contradiction shows that the regularity condition fails at the intersection. In the vector-field construction, the propagation component is generated from these gradients; the loss of regularity leads to a zero propagation direction at the crossing. Hence a singular point is unavoidable. $\square$

The engineering consequence is severe: a vehicle may arrive at the intersection with no unique guidance direction, even though the desired path itself passes through that point.

3.2 Simple Closed Curves

A simple closed curve presents a different but equally fundamental difficulty. Even if the curve is smooth and non-self-intersecting, an autonomous time-invariant vector field cannot make the closed curve globally attractive in the same-dimensional state space.

The paper formalizes this through a topological argument and shows that the domain of attraction cannot be the entire Euclidean space.

Theorem 3.2 (Impossibility of global same-dimensional convergence). *Let $\mathcal{P} \subset \mathbb{R}^n$ be a simple closed or self-intersected desired path described by a conventional same-dimensional guiding vector field. Global convergence of all initial conditions in $\mathbb{R}^n$ to $\mathcal{P}$ cannot be guaranteed; equivalently, the domain of attraction of $\mathcal{P}$ cannot be all of $\mathbb{R}^n$.*

Proof. For a self-intersected path, the result follows from the preceding proposition because at least one singular point lies on the desired path. A trajectory initialized at that singular point cannot be assigned a nonzero path-following direction, so the VF-PF problem itself fails.

For a simple closed path, the result follows from the topological obstruction established in the reference work: a time-invariant autonomous field admitting the desired closed curve as an invariant attracting set cannot make that closed curve globally attracting over the entire Euclidean space without the associated topological consequences that contradict the required global behavior. Equivalently, the phase portrait necessarily contains an obstruction to global attraction. The paper formalizes this argument as its central impossibility theorem. [1] $\square$

The theorem is important because it changes the engineering question. Instead of seeking increasingly complicated gains or local modifications in the physical plane, the problem should be reformulated so that the topology of the desired path is changed.

3.3 Why Normalization Does Not Fix the Problem

Suppose a nonzero field is scaled by a positive state-dependent factor:

$$\tilde{\chi}(\xi) = s(\xi) \frac{\chi(\xi)}{\|\chi(\xi)\|}, \quad s(\xi) > 0. \quad (3.2)$$

Away from the singular set, such scaling changes the speed along integral curves but not their phase portrait. Therefore, normalization cannot remove the underlying topological obstruction. It merely makes the singular points more explicit because the normalized expression is undefined there [1].

Chapter 4

Higher-Dimensional Singularity-Free Formulation

4.1 Core Idea

The reference solution is to introduce an additional coordinate and “lift” the desired path into a higher-dimensional space.

For a planar parameterized path

$$P_{\text{phy}}(w) = \begin{bmatrix} f_1(w) \\ f_2(w) \end{bmatrix}, \quad (4.1)$$

the internship implementation uses

$$\boxed{\gamma(w) = \begin{bmatrix} f_1(w) \\ f_2(w) \\ w \end{bmatrix}} \in \mathbb{R}^3. \quad (4.2)$$

The augmented state is

$$\boldsymbol{\xi} = \begin{bmatrix} x \\ y \\ w \end{bmatrix}. \quad (4.3)$$

A point repeated physically can now occur at two distinct augmented states:

$$(x, y, w_1) \neq (x, y, w_2), \quad w_1 \neq w_2. \quad (4.4)$$

Thus the virtual coordinate stores branch/phase information that is unavailable from (x, y) alone.

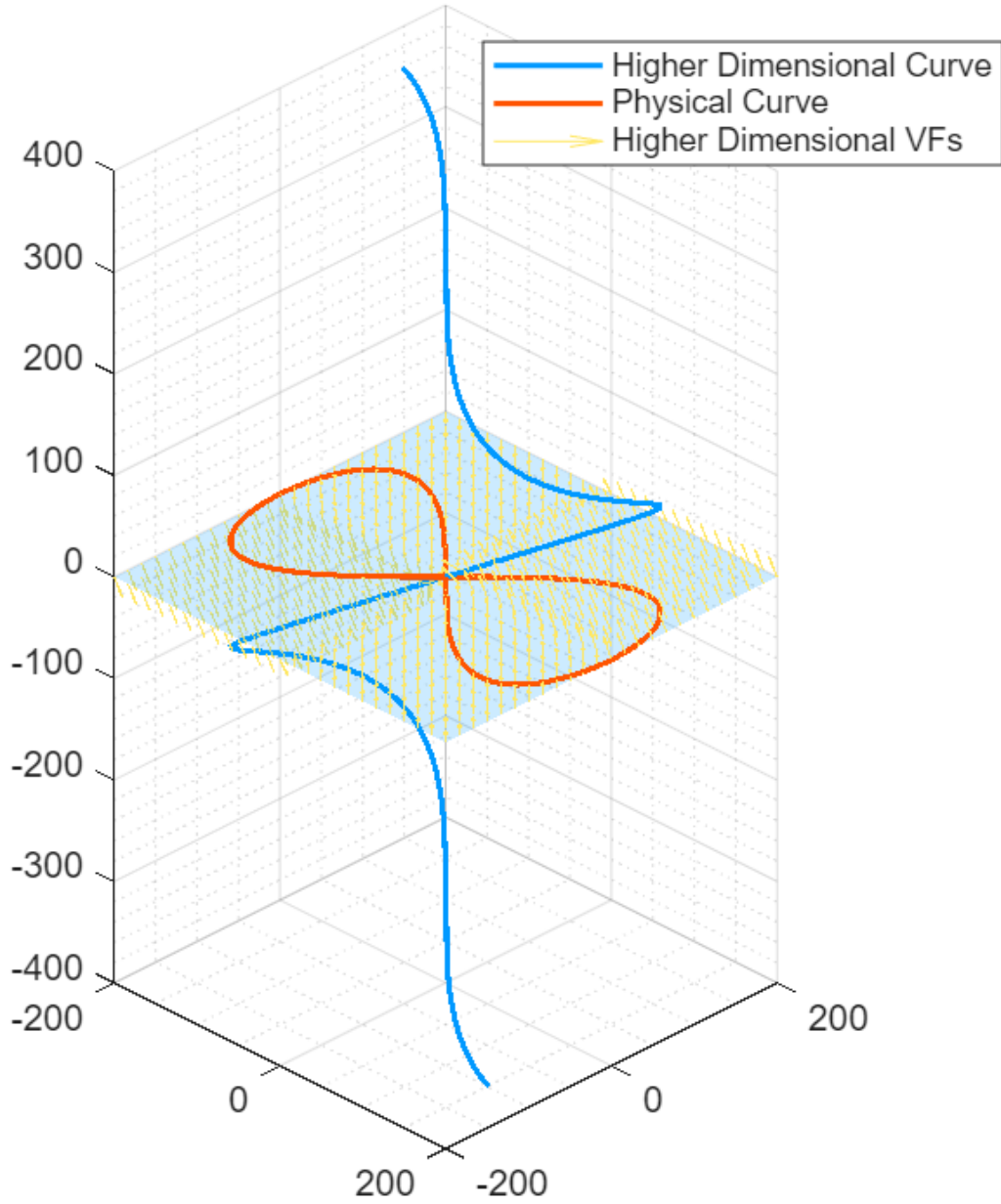


Figure 4.1: Physical and higher-dimensional representation of the figure-eight path.

4.2 Implicit Representation of the Lifted Path

For

$$x = f_1(w), \quad y = f_2(w), \quad (4.5)$$

the project defines

$$\phi_1(x, y, w) = x - f_1(w), \quad (4.6)$$

$$\phi_2(x, y, w) = y - f_2(w). \quad (4.7)$$

Hence

$$\gamma(w) = \{\xi : \phi_1(\xi) = 0, \phi_2(\xi) = 0\}. \quad (4.8)$$

The gradients are

$$\nabla\phi_1 = \begin{bmatrix} 1 \\ 0 \\ -f'_1(w) \end{bmatrix}, \quad \nabla\phi_2 = \begin{bmatrix} 0 \\ 1 \\ -f'_2(w) \end{bmatrix}. \quad (4.9)$$

Their generalized cross product is, up to an orientation sign,

$$\nabla\phi_1 \times \nabla\phi_2 = \begin{bmatrix} f'_1(w) \\ f'_2(w) \\ 1 \end{bmatrix}. \quad (4.10)$$

Most importantly, its third component is identically 1 (or -1 under the opposite orientation). This simple observation is the key to the singularity-free construction.

4.3 Singularity-Free Vector Field

Using the general structure (2.9), the augmented field becomes

$$\chi_{\text{hgh}} = \nabla\phi_1 \times \nabla\phi_2 - K_1\phi_1\nabla\phi_1 - K_2\phi_2\nabla\phi_2, \quad (4.11)$$

for positive gains K_1, K_2 .

Substituting (4.9) and (4.10),

$$\chi_{\text{hgh}} = \begin{bmatrix} f'_1 \\ f'_2 \\ 1 \end{bmatrix} - K_1\phi_1 \begin{bmatrix} 1 \\ 0 \\ -f'_1 \end{bmatrix} - K_2\phi_2 \begin{bmatrix} 0 \\ 1 \\ -f'_2 \end{bmatrix}. \quad (4.12)$$

The third component therefore contains

$$\chi_{\text{hgh},w} = 1 + K_1\phi_1 f'_1(w) + K_2\phi_2 f'_2(w). \quad (4.13)$$

However, the stronger geometric argument is obtained from the orthogonality of the propagation term to both gradients. The propagation vector is never zero because of its constant virtual-coordinate component.

Proposition 4.1 (No singular point on the lifted construction). *Under the regularity assumptions required by the higher-dimensional construction, the vector field obtained from the lifted parameterized path is free of singular points.*

Proof. Let

$$p(\xi) = \nabla\phi_1(\xi) \times \nabla\phi_2(\xi). \quad (4.14)$$

From (4.10), the virtual-coordinate component of p is a nonzero constant. The convergence term

$$q(\xi) = -K_1\phi_1\nabla\phi_1 - K_2\phi_2\nabla\phi_2 \quad (4.15)$$

belongs to the span of $\nabla\phi_1$ and $\nabla\phi_2$, whereas p is orthogonal to this span. Therefore p is orthogonal to q . If

$$p + q = 0, \quad (4.16)$$

then taking the inner product with p gives

$$\|p\|^2 = 0, \quad (4.17)$$

contradicting the fact that the virtual component of p is nonzero. Hence

$$\chi_{\text{hgh}}(\xi) \neq 0 \quad \forall \xi \quad (4.18)$$

within the stated smooth construction. $\square$

This is the central mathematical advantage of the lifted representation: the additional coordinate creates a direction that cannot be cancelled by the convergence term.

4.4 Transformation and Projection

The reference paper formulates a more general transformation operator G_l and proves that convergence in the original augmented dynamics is inherited by the transformed dynamics under appropriate smoothness and Lipschitz conditions [1].

For the linear case, the paper uses

$$P_a = I - \hat{a}\hat{a}^T, \quad \hat{a} = \frac{a}{\|a\|}, \quad (4.19)$$

which projects vectors onto the hyperplane orthogonal to a . Choosing the added-coordinate direction as the transformation direction removes the additional component when interpreting the result in physical coordinates.

The project implementation uses the more direct (x, y, w) representation and treats (x, y) as the physical projection.

4.5 Global Convergence Result

The reference paper establishes that the lifted desired path is unbounded and non-self-intersecting and then derives global asymptotic convergence of the transformed trajectory when the singular set is empty [1].

Theorem 4.2 (Higher-dimensional global convergence). *Let a planar parameterized physical path be lifted to*

$$\gamma(w) = [f_1(w), f_2(w), w]^T, \quad (4.20)$$

and let a corresponding singularity-free higher-dimensional guiding field be constructed under the regularity assumptions of the reference framework. Then the augmented trajectory globally converges to the lifted path, and its physical projection converges to the physical path.

Proof sketch following the reference analysis.

The lifted path is unbounded and non-self-intersecting in the augmented space. Because the higher-dimensional vector field has no singular points, the undesirable singular invariant set is empty. The reference convergence result then leaves the lifted desired path as the only invariant attracting set of the corresponding path-following dynamics. For the transformation operator G_l , global Lipschitz continuity gives

$$\text{dist}(G_l(\xi), G_l(\mathcal{P})) \leq L \text{dist}(\xi, \mathcal{P}), \quad (4.21)$$

where L is a Lipschitz constant. Thus

$$\text{dist}(\xi(t), \mathcal{P}) \rightarrow 0 \implies \text{dist}(G_l(\xi(t)), G_l(\mathcal{P})) \rightarrow 0. \quad (4.22)$$

The coordinate projection of the transformed lifted path reproduces the physical path, so convergence of the augmented trajectory implies convergence of the physical coordinates to the desired physical curve. $\square$

4.6 Exponential Error Convergence

The reference analysis further establishes local or global exponential convergence of the path-following error under bounded derivative conditions on the parameterized path. Let

$$e = \begin{bmatrix} \phi_1 & \phi_2 \end{bmatrix}^T. \quad (4.23)$$

A natural Lyapunov candidate is

$$V(e) = \frac{1}{2} e^T K e. \quad (4.24)$$

Because the propagation term is orthogonal to the constraint gradients, its contribution to the evolution of the constraint error vanishes. The remaining contribution is negative definite in e under the regularity assumptions. The paper uses this structure to establish exponential decay when the required path derivatives are bounded [1].

For the internship report, it is important to distinguish this theoretical result from the numerical UAV guidance layer: the theorem applies to the mathematical vector-field dynamics, not automatically to the finite-turn-rate vehicle coupled to the projected field.

Chapter 5

Internship Implementation

5.1 Project Architecture

The MATLAB implementation follows the sequence

$$\begin{array}{l} \text{Parameterized path} \rightarrow \text{Lifted path} \rightarrow \text{Guiding VF} \\ \rightarrow \text{Physical projection} \rightarrow \text{UAV guidance} \rightarrow \text{Vehicle update} \end{array} \quad (5.1)$$

with visualization and diagnostics around the simulation loop.

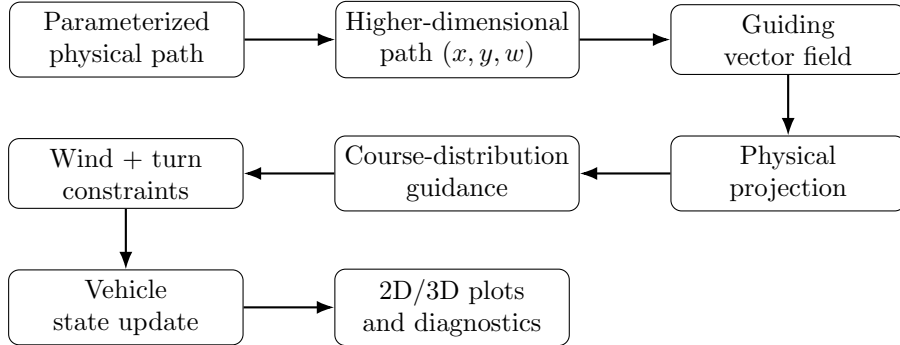


Figure 5.1: High-level architecture of the internship implementation.

5.2 Parameterization of the Path

The project is built around

$$x = f_1(w), \quad y = f_2(w). \quad (5.2)$$

This representation is intentionally generic: the same path functions are used for lifted-path construction, symbolic differentiation, vector-field generation, virtual-parameter initialization, simulation, and visualization.

The principal demonstration case is a figure-eight curve:

$$x(w) = a \sin(\beta w), \quad (5.3)$$

$$y(w) = a \sin(\beta w) \cos(\beta w), \quad (5.4)$$

with lifted curve

$$\gamma(w) = \begin{bmatrix} a \sin(\beta w) \\ a \sin(\beta w) \cos(\beta w) \\ w \end{bmatrix}. \quad (5.5)$$

The figure-eight is especially important because it makes the topology problem visually obvious: the physical curve crosses itself, while the lifted curve separates the two branches along w .

5.3 Smooth Nonlinear Vector-Field Implementation

A project-specific implementation decision was to use smooth nonlinear functions in the practical field evaluation. The code uses terms based on

$$\tanh(\kappa_i \phi_i) \quad (5.6)$$

for convergence behavior and

$$\operatorname{sech}(\kappa_1 \phi_1) \operatorname{sech}(\kappa_2 \phi_2) \quad (5.7)$$

to smoothly modulate the propagation component.

This differs from treating the theoretical formula only as a symbolic expression. The practical implementation is designed to avoid excessively aggressive corrections far from the desired path and to provide bounded, smooth numerical behavior.

It should be emphasized that the singularity-free theorem remains inherited from the higher-dimensional construction. The smooth nonlinear implementation is a project-level realization/adaptation and should not be confused with a new topological theorem.

5.4 Virtual-Parameter Initialization

A major practical question is: given an initial physical position (x_0, y_0) , what should the initial virtual coordinate w_0 be?

The project minimizes the squared physical distance

$$d^2(w) = (f_1(w) - x_0)^2 + (f_2(w) - y_0)^2. \quad (5.8)$$

Stationary points satisfy

$$\frac{d}{dw} d^2(w) = 0. \quad (5.9)$$

The implementation searches for roots across a prescribed parameter interval, applies a second-order admissibility check, evaluates the corresponding distances, and selects the nearest valid candidate.

This is particularly important near self-intersections, where two different values of w can correspond to the same or nearby physical positions. An arbitrary initial value of w can therefore assign the vehicle to the wrong branch of the lifted path.

5.5 Adaptive Chebyshev-Based Root Solver

The root-finding utility developed in the project is designed for nonlinear scalar equations on a finite interval. The computational sequence is:

1. Map the interval to $[-1, 1]$.
2. Evaluate the target function at Chebyshev points.
3. Construct a Chebyshev approximation.
4. Estimate approximation error.
5. Increase approximation degree if required.
6. Perform a midpoint consistency test.
7. Subdivide the interval recursively when necessary.
8. Construct a colleague matrix for the polynomial approximation.
9. Extract candidate roots from its eigenvalues.
10. Filter approximately real roots.
11. Map the roots to the original interval.
12. Refine candidate roots using a numerical local solver.

This implementation is contained in `chebyRoots.m` and provides a more controlled global search over the parameter interval than a single local nonlinear solve.

5.6 Physical Projection

The augmented field is

$$\mathbf{V} = \begin{bmatrix} V_x \\ V_y \\ V_w \end{bmatrix}. \quad (5.10)$$

A fixed-wing-style planar guidance layer can use

$$\mathbf{V}_{\text{phy}} = \begin{bmatrix} V_x \\ V_y \end{bmatrix}, \quad (5.11)$$

and, when $\|\mathbf{V}_{\text{phy}}\|$ is sufficiently large, compute

$$\chi_d = \text{atan2}(V_y, V_x). \quad (5.12)$$

The important practical observation made during the project is

$$\boxed{\|\mathbf{V}\| > 0 \not\Rightarrow \|\mathbf{V}_{\text{phy}}\| > 0}. \quad (5.13)$$

The augmented field may therefore be mathematically valid while the physical XY projection is momentarily unable to define a course.

5.7 Physical-Projection Degeneracy Handling

The implementation explicitly checks

$$\|\mathbf{V}_{\text{phy}}\| < \epsilon. \quad (5.14)$$

Instead of attempting to normalize or pass an ill-conditioned atan2 result into the guidance controller, it holds the current course.

This is a pragmatic numerical safeguard. It is not a proof that the projected field is everywhere nondegenerate, and it does not remove the underlying physical projection issue. The distinction is a central practical insight of the internship.

A useful formal object for future analysis is

$$\mathcal{D} = \{(x, y, w) : V_x = 0, V_y = 0, V_w \neq 0\}. \quad (5.15)$$

5.8 Course-Distribution Guidance

When the desired physical direction is well defined,

$$\chi_d = \text{atan2}(V_{d,y}, V_{d,x}). \quad (5.16)$$

The current course is

$$\chi = \text{atan2}(V_y, V_x), \quad (5.17)$$

and the wrapped course error is

$$e_\chi = \text{wrapToPi}(\chi_d - \chi). \quad (5.18)$$

A proportional course-rate demand is then

$$\dot{\chi}_{cmd} = k_\chi e_\chi. \quad (5.19)$$

This guidance layer serves as an interface between the mathematical vector field and a simplified vehicle model.

5.9 Wind Model

The simulation uses a prescribed airspeed rather than assuming that airspeed and ground speed are identical.

The basic relation is

$$\mathbf{V}_g = \mathbf{V}_a + \mathbf{W}. \quad (5.20)$$

For the current simulation setup:

Table 5.1: Representative simulation configuration.

Parameter	Value
Time step	0.001 s
Simulation duration	150 s
Airspeed	15 m/s
Wind	(3, 4) m/s
Gravity	9.81 m/s ²
Maximum bank angle	$\pi/4$
Course gain	10 s ⁻¹
VF gains	(0.1, 0.1)
Initial position	(200, -200)
Active path	Figure-eight

These are simulation settings rather than universal algorithmic requirements.

5.10 Course-Rate and Curvature Constraints

A geometric field can demand arbitrarily fast heading changes, but a fixed-wing vehicle has a finite turning capability.

Under a simplified coordinated-turn relation,

$$\boxed{|\dot{\chi}| \leq \frac{g \tan \phi_{\max}}{V_a}} \quad (5.21)$$

and the commanded course rate is saturated accordingly.

The corresponding planar curvature estimate is

$$\kappa_{xy} = \frac{\dot{\chi}}{V_g}. \quad (5.22)$$

This provides a direct engineering diagnostic: a geometrically valid vector field is not automatically a physically feasible command for a finite-turn-rate aircraft.

Chapter 6

Simulation and Numerical Framework

6.1 State Representation

The simulation state is defined to capture both the physical motion of the vehicle and its evolution relative to the augmented path. The primary state variables are

$$\mathbf{x}_s = [x, y, w, \dot{x}, \dot{y}, \dot{w}, \ddot{x}, \ddot{y}, \ddot{w}, V_g, \kappa_{xy}, r], \quad (6.1)$$

where x and y denote the physical position of the vehicle, w is the virtual path parameter, V_g is the ground-speed magnitude, κ_{xy} is the instantaneous curvature in the physical XY plane, and r denotes the distance between the vehicle and the current virtual target point (VTP).

The simulation also stores the guidance variables

$$[\omega_x, \omega_y, \omega_z, \chi, \dot{\chi}], \quad (6.2)$$

where χ represents the vehicle course angle and $\dot{\chi}$ is the corresponding course rate. These quantities are used not only for visualization but also for evaluating the physical feasibility of the guidance commands generated by the vector field.

In particular, the inclusion of the course rate and curvature allows the geometric behavior of the guiding vector field to be examined together with the finite turning capability imposed on the vehicle.

6.2 Simulation Configuration

The numerical experiments were performed using the MATLAB implementation developed during the internship. The primary simulation considers a self-intersecting figure-eight path in the physical XY plane and its corresponding lifted representation in the augmented (x, y, w) space.

The principal simulation parameters are listed in Table 6.1.

Table 6.1: Principal simulation parameters.

Parameter	Value
Simulation time step	0.001 s
Simulation duration	150 s
Controller	Dist
Prescribed airspeed	15 m/s
Wind velocity	[3, 4] m/s
Gravitational acceleration	9.81 m/s ²
Maximum bank angle	$\pi/4$
Course gain	10 s ⁻¹
Vector-field gains	[0.1, 0.1]
Initial position	(200, -200) m
Desired path	Figure-eight

The prescribed airspeed is kept constant, while the wind field modifies the resulting ground velocity according to

$$\mathbf{V}_g = \mathbf{V}_a + \mathbf{W}. \quad (6.3)$$

The maximum bank angle is used to impose a finite course-rate limit on the guidance command according to

$$|\dot{\chi}| \leq \frac{g \tan(\phi_{\max})}{V_a}. \quad (6.4)$$

Thus, the simulation does not assume that the vehicle can instantaneously follow an arbitrary direction specified by the mathematical vector field.

6.3 Simulation Loop

The complete simulation procedure consists of the sequential evaluation of the guiding vector field, its physical projection, the course guidance law, and the vehicle-state update.

The procedure is summarized in Algorithm 1.

Algorithm 1 High-level simulation procedure

```

1: Define the parameterized physical path  $f_1(w), f_2(w)$ .
2: Construct the higher-dimensional path  $\gamma(w)$ .
3: Define the implicit path constraints  $\phi_1, \phi_2$ .
4: Generate the numerical guiding-field function.
5: Determine the initial virtual parameter  $w_0$  using distance minimization and root finding.
6: Initialize the vehicle position, velocity, and course.
7: for each simulation time step do
8:   Evaluate the augmented vector field  $\mathbf{V}(x, y, w)$ .
9:   Extract the physical projection  $\mathbf{V}_{\text{phy}} = [V_x, V_y]^T$ .
10:  if  $\|\mathbf{V}_{\text{phy}}\| \geq \epsilon$  then
11:    Compute the desired course  $\chi_d$ .
12:    Compute the wrapped course error  $e_\chi$ .
13:    Generate  $\dot{\chi}_{cmd}$ .
14:    Apply the course-rate constraint.
15:  else
16:    Hold the current course.
17:  end if
18:  Compute the wind-dependent ground velocity.
19:  Update the physical vehicle state and virtual coordinate.
20:  Evaluate curvature and distance to the virtual target.
21:  Store the guidance and vehicle variables.
22:  Update the physical and augmented trajectory visualizations.
23: end for

```

The simulation therefore represents the complete chain from the geometric description of the desired path to the vehicle-oriented guidance command and subsequent state evolution.

6.4 Path-Following Results

The primary simulation uses a self-intersecting figure-eight path. This path is particularly useful for evaluating the proposed formulation because the two branches intersect in the physical plane while remaining distinct in the augmented space due to their different values of the virtual coordinate w .

The resulting physical and augmented trajectories are shown in Figure 6.1.

As seen in the left panel of Figure 6.1, the vehicle starts from the initial position $(200, -200)$ and converges toward the desired figure-eight path. After the initial convergence transient, the actual trajectory closely follows the desired path, including the self-intersecting region.

The result is particularly significant at the intersection of the two branches. In the physical (x, y) plane, both branches occupy the same spatial location. Consequently, a conventional two-dimensional representation cannot distinguish which branch of the path the vehicle is following. In contrast, the right panel shows that these branches are separated along the virtual coordinate w in the augmented space.

Thus, the simulation provides a direct geometric interpretation of the higher-dimensional formulation: the additional coordinate preserves information about the progression along the parameterized path and removes the ambiguity that exists at the physical crossing.

The observed behavior is consistent with the intended role of the higher-dimensional singularity-

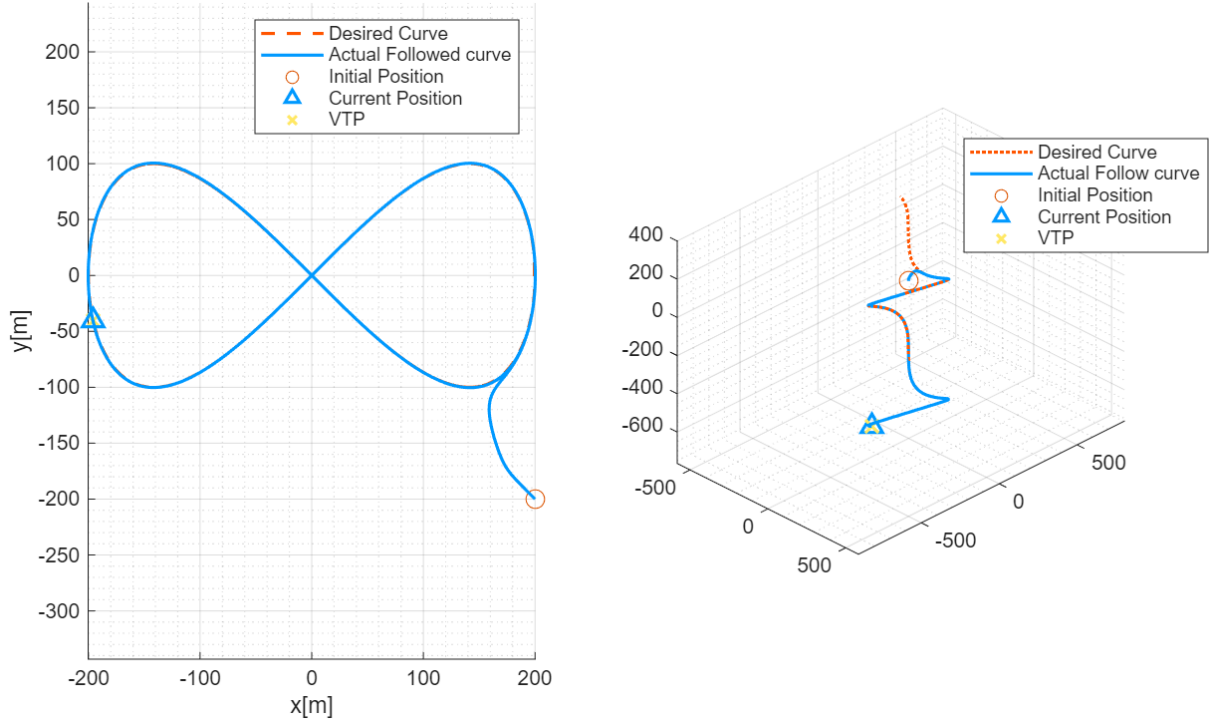


Figure 6.1: Path-following simulation for the self-intersecting figure-eight path. The left panel shows the desired and actual trajectories in the physical (x, y) plane, while the right panel shows the corresponding trajectory in the augmented (x, y, w) space.

free construction, in which the physical path is represented by an unbounded curve in the augmented state space.

6.5 Guidance and Vehicle Diagnostics

While the physical trajectory demonstrates geometric path following, the guidance variables must also be examined to determine whether the commanded motion remains compatible with the assumed vehicle limitations. The principal diagnostic variables are shown in Figure 6.2.

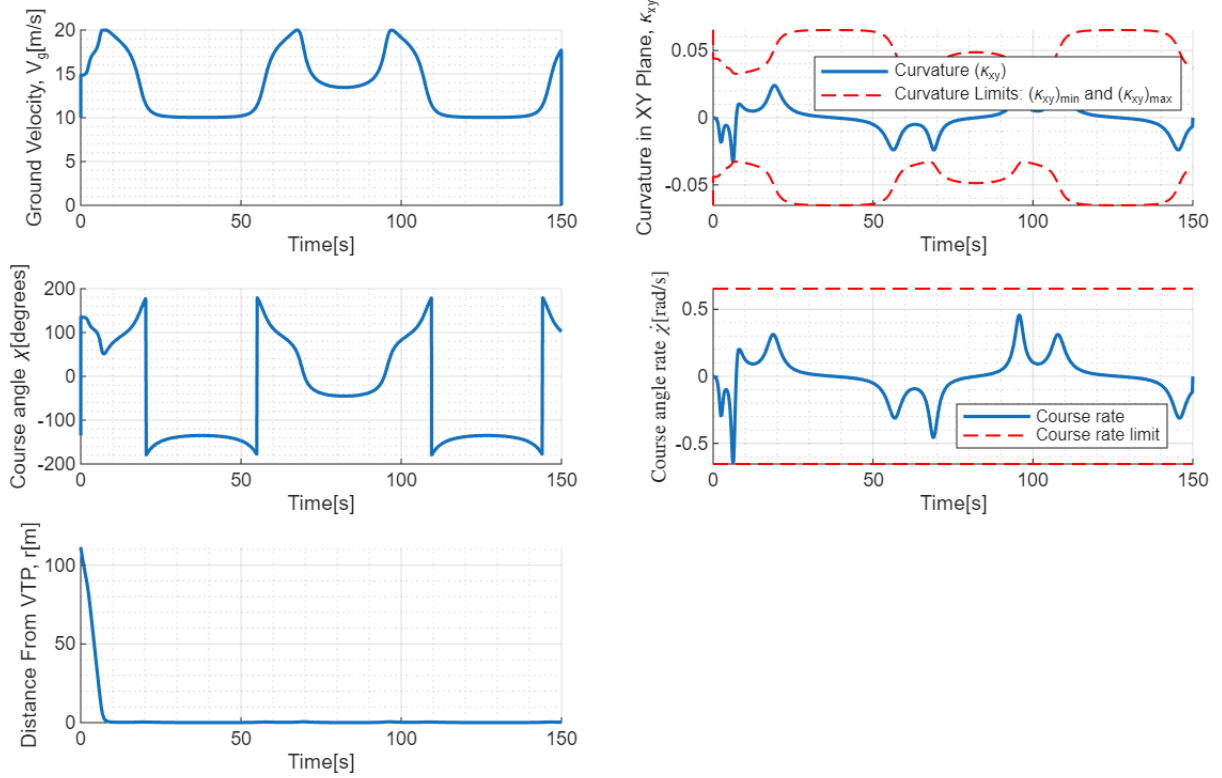


Figure 6.2: Guidance and vehicle-state diagnostics for the path-following simulation. The plots show ground velocity, horizontal-plane curvature, course angle, course rate, and distance from the virtual target over the simulation time.

6.5.1 Ground Velocity

The ground velocity varies throughout the simulation despite the prescribed constant airspeed. This behavior is a direct consequence of the wind model,

$$\mathbf{V}_g = \mathbf{V}_a + \mathbf{W}, \quad (6.5)$$

because the ground-relative velocity depends on the instantaneous vehicle course relative to the wind direction. The variation in ground speed shown in Figure 6.2 is therefore an expected feature of the wind-aware guidance model rather than a variation in the prescribed airspeed.

6.5.2 Horizontal-Plane Curvature

The upper-right plot in Figure 6.2 shows the horizontal-plane curvature κ_{xy} together with the corresponding allowable curvature bounds.

The curvature remains within the prescribed envelope throughout the simulation. This indicates that the direction changes generated by the guidance law remain compatible with the finite turning capability represented by the imposed course-rate constraint.

This is an important distinction between the mathematical vector field and the vehicle-oriented implementation: a guiding field may possess a valid direction at every point while still demanding turns that are too aggressive for a physical vehicle. Monitoring curvature provides a direct means of detecting such situations.

6.5.3 Course Angle

The course-angle history exhibits transitions between approximately -180° and $+180^\circ$. These apparent discontinuities correspond to the conventional wrapping of the angular representation rather than to instantaneous physical rotations of the vehicle.

The wrapped course error used by the controller avoids interpreting these equivalent angular representations as large-magnitude heading errors.

6.5.4 Course Rate

The course-rate history remains within the imposed upper and lower limits throughout the simulation, as shown in the middle-right plot of Figure 6.2. Hence, the commanded course changes remain within the simplified turning capability adopted for the simulation.

This result is important because it demonstrates that the vector-field guidance output has been passed through an explicit vehicle-feasibility constraint rather than being applied directly as an unrestricted direction command.

6.5.5 Distance to the Virtual Target

The bottom-left plot of Figure 6.2 shows the distance r between the vehicle and the virtual target point. Starting from the initial condition, the distance decreases rapidly and subsequently remains close to zero for the remainder of the simulation.

This indicates that the vehicle converges rapidly toward the moving virtual target associated with the parameterized path and subsequently remains close to it. Together with the physical trajectory shown in Figure 6.1, this provides evidence of successful path-following behavior for the simulated configuration.

It should be noted, however, that distance to the virtual target is a guidance diagnostic and should not by itself be interpreted as a complete measure of geometric path error. The physical trajectory and augmented trajectory should be considered together when evaluating the overall path-following performance.

6.6 Discussion of Results

The two sets of results provide complementary evidence regarding the performance of the implemented guidance framework.

First, the physical trajectory demonstrates convergence to and propagation along the desired self-intersecting path. More importantly, the augmented trajectory shows how the virtual coordinate separates branches that overlap in the physical plane.

Second, the quantitative diagnostics demonstrate that the resulting motion is consistent with the simplified vehicle constraints used in the simulation. The ground velocity responds to the wind field, while the curvature and course rate remain within their prescribed limits.

The results therefore demonstrate the complete simulation chain:

Higher-dimensional VF $\rightarrow$ Physical projection $\rightarrow$ Course guidance
 $\rightarrow$ Wind and turn constraints $\rightarrow$ Vehicle trajectory

It is important to emphasize that these results validate the implementation at the guidance and kinematic simulation level. They do not constitute validation of a complete six-degree-of-freedom aircraft controller or experimental flight testing. Such extensions are discussed in the limitations and future-work sections.

6.7 Reproducibility and Software Structure

The implementation is developed in MATLAB and uses the Symbolic Math Toolbox for symbolic path definitions, symbolic differentiation, and generation of the numerical vector-field functions. The Optimization Toolbox is used by utilities such as `fsolve` for solving nonlinear equations.

The complete source code, simulation scripts, numerical utilities, and supporting material are maintained in the following public GitHub repository:

https://github.com/dhe7871/CV_Research_Intern

The repository provides the complete implementation associated with this internship project and can be used to reproduce the simulations and examine the individual components of the guidance framework.

The principal files and modules include:

File/module	Role
<code>main.m</code>	Main simulation and configuration
<code>createVfAtPointMatFunc.m</code>	Numerical vector-field construction/evaluation
<code>initializeCurveParameter.m</code>	Initial virtual-parameter estimation
<code>chebyRoots.m</code>	Adaptive nonlinear root finder
<code>navDistribController.m</code>	Course-distribution guidance
<code>navVelocityController.m</code>	Velocity-oriented guidance utility
<code>plotParams.m</code>	Guidance and vehicle diagnostics
<code>plotVF.m</code>	Vector-field visualization
<code>initializePlotTrajectory.m</code>	Physical and augmented trajectory visualization
<code>normLen.m</code>	Normal-projection utility for implicit curves
<code>phiThetaToUnitVector.m</code>	3D direction utility
<code>Misc/mod_matrix_for_Obs_avoidance.m</code>	Exploratory obstacle-avoidance modulation

The simulation can be executed by entering the MATLAB project directory and running:

```
main
```

The repository also contains supporting reading material and exploratory implementations related to singularity visualization and obstacle avoidance.

Chapter 7

Project-Specific Improvements and Contributions

7.1 Relationship with the Reference Work

The theoretical foundation of this internship is the singularity-free guiding vector-field framework proposed by Yao et al. [1]. The reference work establishes the fundamental topological limitation associated with conventional same-dimensional, time-invariant vector fields for path following. In particular, it demonstrates that global convergence to simple closed and self-intersecting desired paths cannot, in general, be guaranteed without the presence of singular points. To overcome this limitation, the authors introduce a higher-dimensional representation of the desired path, where an additional coordinate is used to transform a potentially closed or self-intersecting physical path into an unbounded and non-self-intersecting curve in an augmented state space.

The formulation implemented during this internship follows this fundamental idea. A parameterized planar path is augmented with an additional virtual coordinate, resulting in a higher-dimensional path on which the guiding vector field can be constructed using the gradients of the corresponding implicit path constraints. The resulting augmented dynamics and their relationship to the physical projection also follow the framework established in [1]. Thus, the singularity-free theorem and the associated topological and convergence results presented in the reference paper are not claimed as new theoretical contributions of this internship. Instead, the internship work focuses on implementing, adapting, and extending the framework for a practical UAV-oriented numerical simulation environment.

7.2 Project-Specific Improvements and Extensions

The implementation developed during the internship introduces several practical components around the theoretical formulation. One of the first modifications concerns the numerical realization of the guiding vector field. The implementation uses bounded smooth nonlinear functions based on tanh and sech for the convergence and propagation terms. This provides a smooth numerical realization of the field and avoids relying solely on an ideal symbolic representation when evaluating the vector field during simulation.

A second practical extension concerns the initialization of the virtual path parameter. Since the augmented state contains the additional coordinate w , an appropriate initial value of w must be determined from the vehicle's initial physical position. Rather than selecting this value arbitrarily, the implementation formulates the initialization problem as a distance-minimization

problem between the initial vehicle position and the parameterized physical curve. Candidate stationary points are identified numerically, examined using a second-order condition, and the closest valid point is selected as the initial virtual target. This procedure becomes particularly important for self-intersecting paths, where multiple parameter values may correspond to the same physical location.

To support this initialization procedure, a dedicated adaptive Chebyshev-based root solver was developed. The solver approximates a nonlinear scalar function over a finite interval using Chebyshev polynomials, evaluates the quality of the approximation, subdivides the interval when necessary, and refines candidate roots numerically. This provides a more systematic approach to finding all relevant roots over the parameter interval than relying only on a single local nonlinear solver.

Another important extension is the incorporation of a practical UAV guidance layer around the higher-dimensional vector field. The theoretical vector field provides a desired direction, but the physical vehicle operates in the planar (x, y) space and cannot instantaneously realize arbitrary changes in direction. The implementation therefore projects the augmented vector field onto the physical plane and converts the projected direction into a desired course command. The vehicle is modeled with a prescribed airspeed while wind is included in the ground-velocity calculation, allowing the interaction between the guidance field and environmental disturbances to be examined.

The guidance layer also incorporates finite turning capability. The desired course-rate command is constrained using a maximum-bank-angle relationship, and the corresponding horizontal-plane curvature is monitored throughout the simulation. This provides an important practical connection between the mathematical guidance field and the physical limitations of an aerial vehicle. A vector field may be mathematically well defined while still demanding course changes that are not realizable by the vehicle. Explicitly monitoring course rate and curvature therefore allows such behavior to be identified during simulation.

An additional observation arising from the implementation is the distinction between singularity of the complete augmented vector field and degeneracy of its physical projection. The higher-dimensional formulation aims to maintain a nonzero augmented vector field,

$$\|\mathbf{V}\| > 0, \quad (7.1)$$

but this does not necessarily imply that the physical projection

$$\mathbf{V}_{\text{phy}} = \begin{bmatrix} V_x \\ V_y \end{bmatrix} \quad (7.2)$$

is nonzero. It is therefore possible to encounter a state satisfying

$$V_x \approx 0, \quad V_y \approx 0, \quad V_w \neq 0, \quad (7.3)$$

where the complete augmented field still has a valid direction but the physical projection cannot directly define a course angle. The implementation explicitly detects this condition and avoids attempting to normalize the nearly zero physical vector. Instead, the current course is retained until a meaningful physical direction becomes available. This handling represents a practical engineering extension of the theoretical formulation rather than a modification of the underlying singularity-free theorem.

The project also includes an extensive visualization and diagnostic framework. The physical trajectory is examined in the conventional (x, y) plane while the corresponding augmented trajectory is visualized in (x, y, w) space. The vector field itself can be examined on selected planes, and the evolution of the virtual target, ground velocity, course angle, course rate,

curvature, and distance to the virtual target can be monitored throughout the simulation. These tools are particularly useful for understanding the geometric effect of the virtual coordinate and for identifying situations in which the mathematical guidance field interacts unfavorably with the physical vehicle constraints.

7.3 Contribution Boundary

The contribution of this internship is therefore best characterized as an implementation-oriented and analysis-oriented extension of an existing singularity-free vector-field theory. The work does not claim the original topological impossibility result or the singularity-free construction as a new theoretical contribution; these are adopted from the reference work of Yao et al. [1]. Instead, the principal contribution lies in developing a practical numerical framework around the theory and studying the additional issues that arise when the higher-dimensional vector field is connected to a UAV-oriented guidance model.

In particular, the internship combines the higher-dimensional vector-field formulation with systematic virtual-parameter initialization, adaptive numerical root finding, wind-aware guidance, finite-turn-rate constraints, curvature monitoring, physical-projection degeneracy handling, and comprehensive physical and augmented-space visualization. These additions turn the theoretical construction into a reproducible simulation framework for investigating singularity-free path following in a UAV guidance context.

The internship also investigated the possibility of incorporating obstacle avoidance into the same framework. Although an exploratory vector-field modulation approach was implemented, it did not produce a satisfactory integrated obstacle-avoidance solution with corresponding convergence and safety guarantees. Consequently, obstacle avoidance is treated in this work as an exploratory extension and a direction for future research rather than as a completed contribution. This distinction is important because modifying the guiding field near obstacles can introduce new equilibria or disrupt the properties of the original singularity-free formulation.

Overall, the work should therefore be viewed as a practical study of how a theoretically singularity-free higher-dimensional vector field can be implemented, numerically evaluated, and coupled to a simplified UAV guidance model, together with an investigation of the additional challenges that must be addressed before the framework can be extended to a fully integrated obstacle-aware autonomous navigation system.

Chapter 8

Results and Technical Observations

8.1 Singularity-Free Augmented Path Following

The central technical result is the successful implementation of the higher-dimensional path-following architecture for parameterized planar curves, including a self-intersecting figure-eight example.

The important geometric observation is that the physical crossing is no longer a single state in the augmented space:

$$(x, y, w_1) \neq (x, y, w_2), \quad w_1 \neq w_2. \quad (8.1)$$

Thus the vector field retains branch/phase information while its physical projection still traces the desired planar path.

8.2 Physical Projection Is a Separate Engineering Problem

One of the clearest practical conclusions is that “singularity-free” must be interpreted carefully.

The mathematical guarantee concerns

$$\mathbf{V} \neq 0. \quad (8.2)$$

A real planar vehicle, however, needs

$$\mathbf{V}_{\text{phy}} \neq 0 \quad (8.3)$$

to extract a course directly.

Therefore,

$$\|\mathbf{V}\| > 0 \quad \text{does not guarantee} \quad \|\mathbf{V}_{\text{phy}}\| > 0. \quad (8.4)$$

This is not a contradiction of the singularity-free theorem. It arises because projection discards the very virtual-coordinate direction that makes the augmented vector field robust against cancellation.

The project handles this through a tolerance and course hold, but this is a fallback rather than a complete analytical solution.

8.3 Vehicle Feasibility Matters

The ideal vector field specifies direction continuously, but the vehicle can only realize directions at a finite rate. The course-rate saturation in (5.21) makes this limitation explicit.

The diagnostics therefore serve two purposes:

1. assess geometric path-following behavior;
2. assess whether the required motion is compatible with the simplified vehicle turning model.

This distinction is especially important for high-curvature paths, aggressive gains, and strong wind.

8.4 Parameterization Has a Physical Effect

Although different parameterizations can trace the same geometric curve, the augmented formulation contains the parameter itself as a state coordinate. Consequently, changing

$$w = g(s) \tag{8.5}$$

changes the lifted geometry and the evolution of the virtual state.

This means that parameterization is not merely a notational choice once the virtual coordinate is used as part of the dynamical system. It can influence convergence behavior, curvature, course rate, numerical conditioning and the physical projection.

Chapter 9

Exploratory Obstacle-Avoidance Investigation

9.1 Motivation

A natural next step after obtaining singularity-free path following is to introduce obstacle avoidance so that the vehicle can deviate from the desired path when a collision threat appears and later recover the desired path.

The internship therefore included an exploratory obstacle-avoidance implementation based on local vector-field modulation.

9.2 Obstacle Representation

For an implicit obstacle boundary

$$F(x, y) = 0, \tag{9.1}$$

the outward/normal direction is computed from

$$\mathbf{n} = \frac{\nabla F}{\|\nabla F\|}. \tag{9.2}$$

A planar tangent can be chosen as

$$\mathbf{t} = \begin{bmatrix} -n_y \\ n_x \end{bmatrix}. \tag{9.3}$$

The exploratory module incorporates the following ingredients:

- obstacle gradients;
- normal and tangent directions;
- distance-based influence;
- safety margins;
- a modulation matrix;
- directional rotation;
- spatial decay;

- symmetry breaking in the downstream region.

The implementation is contained separately in `mod.matrix_for.Obs_avoidance.m`.

9.3 Attempted Integration

The intended architecture was conceptually

$$\boxed{\text{Singularity-free path-following field} + \text{Obstacle-modified field}} \quad (9.4)$$

with obstacle influence becoming strong near the obstacle and weak away from it.

The difficulty is that the path-following field and the obstacle-avoidance field satisfy different objectives. The path-following field wants to converge to a one-dimensional geometric manifold, while the obstacle field must create a temporary safe detour and subsequently recover a direction that leads back to the desired path.

9.4 Outcome

The obstacle-avoidance part of the internship did not result in a satisfactory integrated algorithm. In particular, the exploratory modulation was not elevated to the status of a validated guidance law with a corresponding proof of simultaneous collision avoidance and path convergence.

This is an important negative result and should be reported explicitly rather than presenting the obstacle module as completed.

The current state of the work is therefore:

Table 9.1: Status of the main research components.

Component	Status
Higher-dimensional singularity-free VF construction	Implemented
Parameterized 2D path following	Implemented
Self-intersecting figure-eight example	Implemented
Virtual-parameter initialization	Implemented
Chebyshev root solver	Implemented
Wind-aware guidance	Implemented
Course-rate/curvature constraints	Implemented
Physical-projection degeneracy handling	Implemented
2D/3D visualization and diagnostics	Implemented
Obstacle modulation	Exploratory
Fully integrated obstacle avoidance with convergence/safety proof	Not achieved

9.5 Research Lessons from the Unsuccessful Extension

The failed integration is technically informative.

First, avoiding a collision is not enough: the modified field must still possess a reliable escape mechanism and must reconnect to the desired path.

Second, adding vector fields or rotations can create new equilibria or poorly conditioned directions, even when each component field is individually well behaved.

Third, the singularity-free guarantee of the lifted path-following field cannot automatically be transferred to an arbitrarily modulated obstacle field. Once the vector field is modified, its nonvanishing property must be re-established for the combined field.

Fourth, collision avoidance and path convergence should ideally be treated as a coupled dynamical-systems problem rather than only as a local vector addition or direction blending problem.

These points motivate the future research directions presented later in the report.

Chapter 10

Limitations

10.1 Guidance-Level Vehicle Model

The simulation does not represent a complete six-degree-of-freedom rigid-body aircraft. In particular, it does not fully include:

- aerodynamic force and moment models;
- roll and pitch dynamics;
- actuator dynamics and saturation;
- sideslip;
- load-factor constraints;
- sensor uncertainty;
- time-varying airspeed.

Therefore, success in the guidance simulation should not be interpreted as flight-level validation.

10.2 Simplified Turning Constraint

Equation (5.21) uses a simplified coordinated-turn relationship. It is useful for making the guidance problem physically meaningful, but it is not a substitute for a full aircraft dynamics and control model.

10.3 Parameter Sensitivity

Trajectory quality depends on vector-field gains, course gain, time step, path parameterization, wind, airspeed and turning constraints. Gain selection can affect convergence speed, oscillations, path error, saturation and curvature.

10.4 Projection Degeneracy

The current course-hold strategy is a practical fallback. It does not establish that the projected guidance field has a unique physically meaningful direction everywhere.

10.5 Obstacle-Avoidance Guarantees

The obstacle-avoidance extension is not a validated collision-avoidance planner. There is currently no complete proof of simultaneous:

1. collision avoidance;
2. recovery to the desired path;
3. absence of newly introduced singularities;
4. and global convergence under the full combined field.

Chapter 11

Future Work

11.1 Analytical Characterization of Projection Degeneracy

The set

$$\mathcal{D} = \{(x, y, w) : V_x = 0, V_y = 0, V_w \neq 0\} \quad (11.1)$$

should be characterized analytically. This would determine whether projection degeneracy is isolated, path-dependent, parameterization-dependent, or structurally unavoidable.

11.2 Predictive Handling of Degenerate Directions

Instead of simply holding the current course, one could use local derivatives of the augmented field to predict the next physically informative direction. Such a method would preserve the vehicle's motion continuity while respecting the mathematical role of the virtual coordinate.

11.3 Parameterization Optimization

The parameterization could be treated as a design variable and optimized for:

$$\begin{array}{ll} \text{convergence rate,} & \text{path-following error,} & \text{curvature,} \\ \text{course rate,} & \text{control effort,} & \text{numerical conditioning.} \end{array} \quad (11.2)$$

This is especially relevant because the virtual coordinate becomes part of the dynamical state.

11.4 Constant-Speed Formulations

The higher-dimensional formulation naturally introduces a virtual-coordinate component whose scaling can alter the speed of the augmented trajectory. Alternative normalization strategies could seek improved constant-speed behavior without creating numerical problems near degenerate physical projections.

11.5 Full Six-Degree-of-Freedom UAV Dynamics

The next validation level should replace the kinematic guidance model with a six-degree-of-freedom rigid-body simulation including:

- position and attitude dynamics;
- aerodynamic forces and moments;
- actuator limitations;
- sensor noise;
- varying wind;
- varying airspeed;
- inner-loop attitude control.

11.6 Integrated Obstacle Avoidance

The obstacle-avoidance problem should be reformulated so that obstacle repulsion and path recovery are properties of one analytically characterized dynamical system. A successful formulation should seek to prove, under explicit assumptions:

$$\text{safe set forward invariance} \tag{11.3}$$

and

$$\text{dist}(\boldsymbol{\xi}(t), \mathcal{P}) \rightarrow 0 \tag{11.4}$$

after the obstacle-induced detour.

The key requirement is to retain a nonvanishing guiding direction while modifying the field near obstacles.

11.7 Three-Dimensional Physical Paths

A natural extension is from

$$(x, y) \tag{11.5}$$

to spatial paths

$$(x, y, z), \tag{11.6}$$

with an additional virtual coordinate. This could directly support 3D UAV inspection, atmospheric sampling and terrain-following problems.

Chapter 12

Conclusion

The research internship investigated a fundamental problem in vector-field-based path following: the inability of conventional same-dimensional, time-invariant vector fields to guarantee globally convergent motion to simple closed and self-intersecting paths without singularities.

The central solution studied in this work was the higher-dimensional formulation introduced by Yao et al. [1]. By augmenting a parameterized planar path

$$[f_1(w), f_2(w)]^T \quad (12.1)$$

with the virtual coordinate w , the difficult physical topology is represented by an unbounded lifted curve

$$[f_1(w), f_2(w), w]^T. \quad (12.2)$$

The resulting guiding vector field contains a propagation direction with a nonzero virtual-coordinate component, preventing complete cancellation by the converging component under the construction assumptions. The associated dynamical-systems analysis then provides global convergence results for the lifted path.

The internship work went beyond reproducing the theoretical construction. It developed a complete MATLAB framework around the method, including numerical virtual-parameter initialization, an adaptive Chebyshev root solver, smooth nonlinear field realization, wind-aware course guidance, finite-turn-rate and curvature constraints, physical-projection degeneracy handling, and synchronized 2D/3D visualization and diagnostics.

A particularly important observation was the distinction

$$\|\mathbf{V}\| > 0 \not\Rightarrow \|\mathbf{V}_{\text{phy}}\| > 0. \quad (12.3)$$

The augmented vector field can remain mathematically valid while its physical projection temporarily fails to define a course. Recognizing and handling this condition is essential when connecting the higher-dimensional mathematical theory to a real vehicle.

The attempted extension to obstacle avoidance was not successful enough to claim a completed integrated solution. The project instead established an exploratory modulation framework and, more importantly, identified the need for a stronger coupled analysis of safety, path recovery and singularity avoidance. This negative result is a legitimate research outcome because it defines the boundary between what is currently validated and what remains open.

Overall, the internship produced a research-oriented simulation framework for singularity-free vector-field path following and clarified several practical issues that emerge when a theoretically singularity-free guidance field is coupled to a vehicle-oriented navigation architecture. The

most direct next step is to develop a mathematically consistent obstacle-aware extension while retaining the singularity-free property and then validate the complete guidance law using full six-degree-of-freedom UAV dynamics and, ultimately, flight experiments.

Appendix A

Key Mathematical Identities

A.1 Lifted Path

$$\gamma(w) = \begin{bmatrix} f_1(w) \\ f_2(w) \\ w \end{bmatrix}. \quad (\text{A.1})$$

A.2 Constraint Functions

$$\phi_1 = x - f_1(w), \quad \phi_2 = y - f_2(w). \quad (\text{A.2})$$

A.3 Constraint Gradients

$$\nabla\phi_1 = \begin{bmatrix} 1 \\ 0 \\ -f'_1(w) \end{bmatrix}, \quad \nabla\phi_2 = \begin{bmatrix} 0 \\ 1 \\ -f'_2(w) \end{bmatrix}. \quad (\text{A.3})$$

A.4 Propagation Direction

$$\nabla\phi_1 \times \nabla\phi_2 = \pm \begin{bmatrix} f'_1(w) \\ f'_2(w) \\ 1 \end{bmatrix}. \quad (\text{A.4})$$

A.5 Generic Higher-Dimensional Field

$$\chi = \nabla \times \phi - NKe. \quad (\text{A.5})$$

A.6 Course Law

$$\chi_d = \text{atan2}(V_{d,y}, V_{d,x}), \quad (\text{A.6})$$

$$e_\chi = \text{wrapToPi}(\chi_d - \chi), \quad (\text{A.7})$$

$$\dot{\chi}_{cmd} = k_\chi e_\chi. \quad (\text{A.8})$$

A.7 Turning Constraint

$$|\dot{\chi}| \leq \frac{g \tan \phi_{\max}}{V_a}. \quad (\text{A.9})$$

A.8 Ground Velocity

$$\mathbf{V}_g = \mathbf{V}_a + \mathbf{W}. \quad (\text{A.10})$$

Bibliography

- [1] W. Yao, H. G. de Marina, B. Lin, and M. Cao, “Singularity-free guiding vector field for robot navigation,” *IEEE Transactions on Robotics*, vol. 37, no. 4, pp. 1206–1221, 2021.
- [2] J. Fernandez Galarreta, N. Kerle, and M. Gerke, “UAV-based urban structural damage assessment using object-based image analysis and semantic reasoning,” *Natural Hazards and Earth System Science*, vol. 15, no. 6, pp. 1087–1101, 2015.
- [3] E. W. Frew, “Cooperative standoff tracking of uncertain moving targets using active robot networks,” in *IEEE International Conference on Robotics and Automation*, (Piscataway, NJ), pp. 3277–3282, IEEE, 2007.
- [4] T. Oliveira, A. P. Aguiar, and P. Encarnacao, “Moving path following for unmanned aerial vehicles with applications to single and multiple target tracking problems,” *IEEE Transactions on Robotics*, vol. 32, no. 5, pp. 1062–1078, 2016.
- [5] F. Remondino, L. Barazzetti, F. Nex, M. Scaioni, and D. Sarazzi, “UAV photogrammetry for mapping and 3D modeling—current status and future perspectives,” *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 38, no. 1, p. C22, 2011.
- [6] S. Park, J. Deyst, and J. How, “A new nonlinear guidance logic for trajectory tracking,” in *AIAA Guidance, Navigation, and Control Conference and Exhibit*, AIAA Paper 2004-4900, 2004.
- [7] I. Kaminer, A. Pascoal, E. Hallberg, and C. Silvestre, “Trajectory tracking for autonomous vehicles: An integrated approach to guidance and control,” *Journal of Guidance, Control, and Dynamics*, vol. 21, no. 1, pp. 29–38, 1998.
- [8] P.-J. Yuan and J.-S. Chern, “Ideal proportional navigation,” *Journal of Guidance, Control, and Dynamics*, vol. 15, no. 5, pp. 1161–1165, 1992.